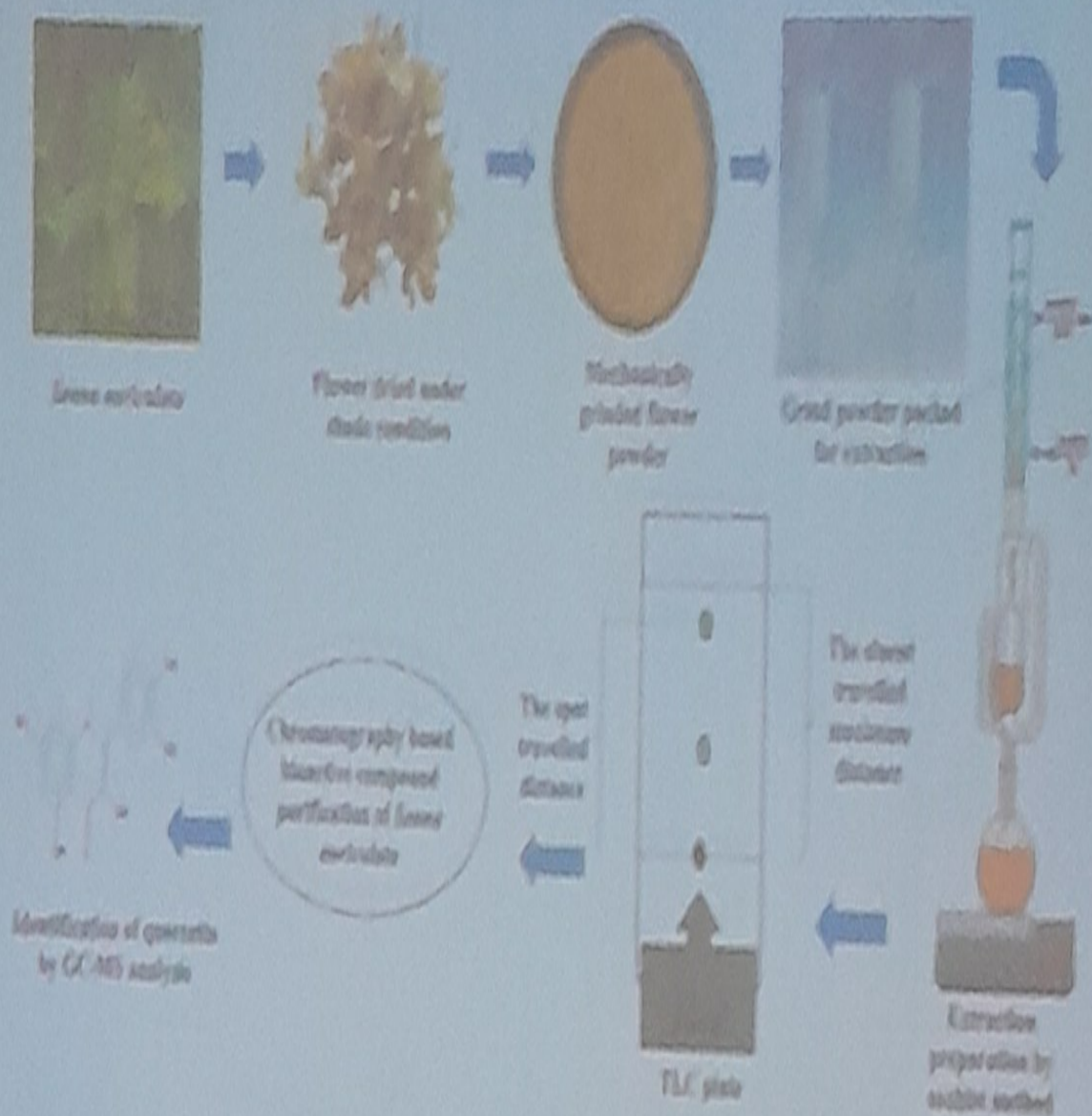


Steps in the scientific analysis of drugs from natural resources



- Basic steps involved are:

- Selection of plant material
- Collection and Taxonomic identification of the plant
- Literature survey on the identified plant
- Design of appropriate extraction and separation methods
- Checking extracts/ fractions for pharmacological activity
- Identification of classes of compounds found in the plant (phytochemical screening)
- Isolation of active compounds or fractions responsible for the pharmacological activity of the plant

1. Selection and collection of plant material

- There are at least five recognized approaches to choose plants and other organisms for the discovery of newer drugs.

A. Random screening

- Using the random approach, plants are collected regardless of any existing previous knowledge of their chemistry or biological activity.
- This approach relies on the availability of plants that are abundant in a certain area.
- This approach is purely serendipitous in that there is a chance that random plant selection will give access to extracts (or compounds) with biological activity (bioactivity).
 - Eg. High-throughput process development (HTPD) of plants in the 1970s

B. Ethnomedical approach

- In this approach, knowledge of the use of a particular plant by an indigenous people is used to direct a search for a drug lead.
 - E.g., Investigating remedies used in traditional medicine by "shamans" or medicine men or women.
- By collecting and reviewing ethnopharmacological/ ethnomedicinal information and data of a certain community or ethnicity
 - E.g. Online data of published ethnopharmacological studies.

C. Selection of specific taxonomic groups

- It narrows down the search to specific families or genera that have produced relevant phytochemicals in drug discovery in the past.
- Selecting a plant with the intent of a specific pharmacological activity.

D. Chemotaxonomic approach

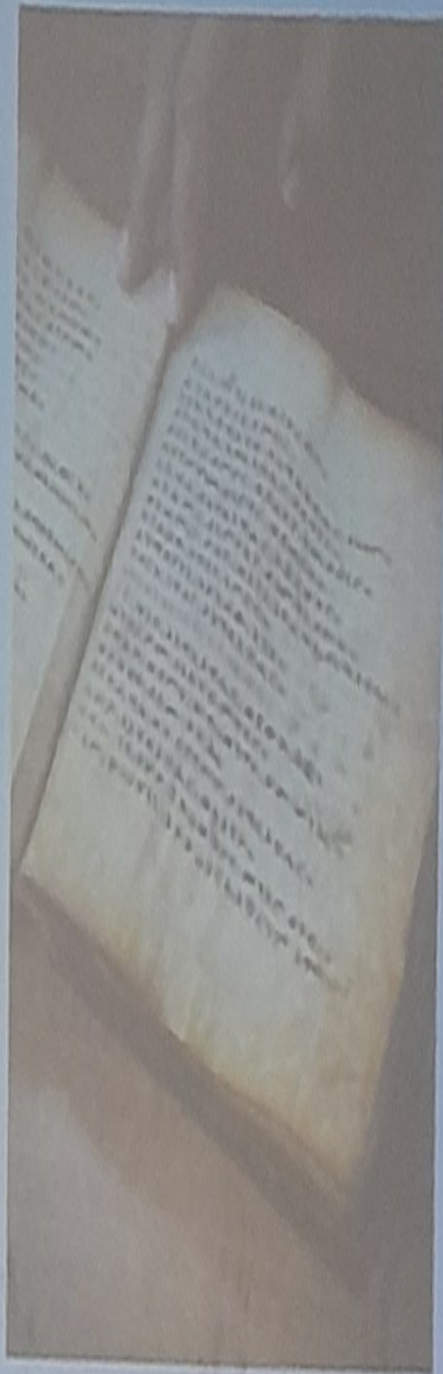
- Way of selection of plant material by focusing on the availability of certain classes of a compound of our interest in a genera or species of a plant
 - such as alkaloids or a flavonoid
- This means that if species/genera related to the plant under investigation are known to contain specific compounds, then the plant itself can be expected to contain similar compounds.
- This approach is highly useful when the chemistry and biological activity of a compound is well described and compounds with similar chemical structure are needed for further biological testing.
 - Eg. plant family Solanaceae, which is a rich source of alkaloids of the tropane type hyoscine.

E. Ancient documents

- Doctrine of signature

- መጽሃፈ ፊውስ

- መጽሃፈ አድህኖ



2. Collection and taxonomic identification of the plant

The whole plant or a particular plant part can be collected depending on where the metabolites of interest (if they are known) accumulate.

- Hence, aerial (e.g., leaves, stems, flowering tops, fruits, seeds, bark) and underground (e.g., bulbs, tubers, roots) parts can be collected separately.
- Only healthy specimens should be obtained, as signs of contamination (fungal, bacterial, or viral) may be linked to a change in the profile of metabolites present.

- It should be stressed that the plant must also be identified correctly.
- Taxonomic authentication is extremely important because plants are often misidentified by folk users and in the literature
- A specialized taxonomist should be involved in the detailed authentication of the plant (i.e., classification into its species, genus, family, order, and class).
- A dried whole plant specimen pressed between sheets of paper deposited in a herbarium for future reference.
 - the name of the plant, the identity of the part(s) collected, the place and date of collection, should be recorded as part of a voucher

3. Literature survey on the identified plant

- It is the process of digging out what has been done in past on the plant we just selected using the internet and other documents

Eg. Google scholar, pubmed, elsevier

- Helps to avoid redundancy of work
- Gives more information on

phytochemistry, medicinal use, hint on how to extract ...

Eg. Search both *Ruta chalepensis*, *Vernonia amygdalina* on the internet

4. Design of appropriate extraction and separation methods

- Extraction is the **removal, withdrawal or separation** of valuable substances from a **biological matrix**.
- In designing an extraction method consider this about the active ingredient:
 - What is the chemistry of active ingredient (if we know it)
 - How was it extracted previously, (how its extracted traditionally)
 - How much of the active ingredient do we need to extract (bulk production)
 - At what purity
- The extraction method should be:
 - Cheap and easily available
 - Effective (exhaustive as possible)
 - Safe to use (chemicals, apparatus)

5. Checking extracts/ fractions for pharmacological activity

- Evaluation of an extract, fraction or pure compound using a biological system is called Bioassay
- Three major types:
 - *In vivo*: Occurs in a living biological organism (antimalarial, diuretic activity)
 - *Ex vivo*: Involves an organ or section of a tissue out side of the living organism (GI motility test in organ bath)
 - *In vitro*: performed at cellular or enzyme level (Eg antimicrobial, anticancer)

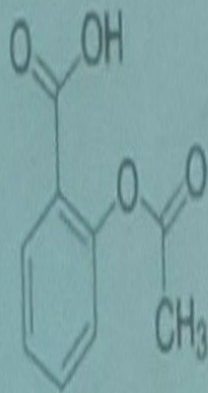
Phytochemical	Name of test	Colour for positive test
1. Flavonoids	Shinoda	Pink
2. Alkaloids	Wagner	Blue-black
3. Tannins	Ferric chloride	Blue-black/Green
4. Terpenoids	Salkowski	Reddish-brown
5. Saponins	Foam test	Foam formation
6. Simple phenols	Ferric chloride	Green
7. Polyphenols	Ferric chloride	Blue
8. Anthocyanins	NaOH	Blue-green
Betacyanins	NaOH	Yellow
Quinones	HCl	Green
Phlobatannins	HCl	Red precipitate
Anthraquinones	HCl+chloroform+ammonia	Rose-pink/violet
Coumarins	NaOH+chloroform	Yellow
Phytosterols	Salkowski	Red
Cardiac glycosides and Cardenolides	Keller-Kiliani's	Brown-red ring
Reducing sugars	Benedict's	Red precipitate
Proteins	Biuret	Violet
Amino acids	Ninhydrin	purple
Fatty acids	Diethyl ether	Transparent stain

7. Isolation of active compounds or fractions responsible for the pharmacological activity of the plant

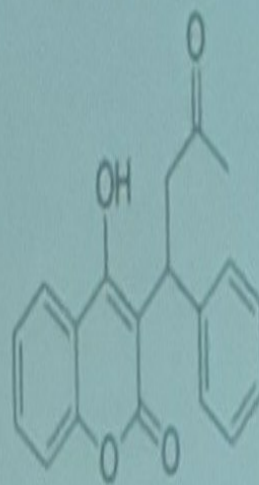
- Narrowing down to a single active compound
 - Total isolation or bioassay guided isolation
- The general features of the molecule that are helpful for the isolation process include polarity (hydrophobicity or hydrophilicity), acid-base properties, chemical stability, and molecular size.
- The most commonly used methods for fractionation and isolation are chromatographic techniques.

8. Structure Elucidation

- It's the process of identifying the chemical structure of the isolated compound



Acetylsalicylic acid (Aspirin)



warfarin (dicumarol derivative)

The following spectroscopic techniques are generally used

Ultraviolet-visible spectroscopy (UV-vis):

- Provides information on chromophores present in the molecule.

Infrared spectroscopy (IR):

- Determines different functional groups, e.g., —C=O , —OH , —NH_2 , aromaticity,

Mass spectrometry (MS):

- Gives information about the molecular mass, molecular formula, and fragmentation pattern.
- electron impact mass spectrometry (EIMS), chemical ionization mass spectrometry (CIMS),
- electrospray ionization mass spectrometry (ESIMS), and fast atom bombardment mass spectrometry (FABMS).

Nuclear magnetic resonance spectroscopy (NMR):

- Reveals information on the number and types of protons and carbons (and other elements like nitrogen, fluorine, etc.) present in the molecule, and the relationships among these atoms.

